SOLUTIONS

1. (a)

Each correct matching counts 2 marks (12 marks on total).

The function $f(x_1, x_2)$ is black over the square shown. To make the matches, you have to ask yourself what set of points will correspond to those points under the given transformation of f. The matches are as follows:

1. $f(ax_1, x_2)$: Since $a > 1$ this shrinks the square in the x_1-direction and so that's figure (vi), i.e., the rectangle in figure (vi) will be stretched to the square via $(x_1, x_2) \mapsto (ax_1, x_2)$ and so $f(ax_1, x_2)$ will be black over the rectangle in (vi). For the Fourier transform, $\mathcal{F}(f(ax_1, x_2) = (1/a)\mathcal{F}f(\xi_1/a, \xi_2)$, so that's *stretched* in the ξ_1 direction and the whole figure is likewise stretched. This matches with figure (A).
2. $f(x_1, ax_2)$. Here the reasoning is the same as in the previous part, but applied in the x_2 and ξ_2 directions. The matches are with (ii) and (F).
3. $f(x_1 + a, x_2)$. This is a shift in the x_1 direction by an amount a to the left, so the square hasn't changed shape, just location relative to where it was before. There is only a phase change in the Fourier transform, which has magnitude 1, thus the plot is the same as for $\mathcal{F}f(\xi_1, \xi_2)$. The matches are with (v) and (C).
4. $f\left(A\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)$. The matrix A is a counterclockwise rotation through $\pi/6$. As we showed in class, the spectrum is also rotated counterclockwise by $\pi/6$. Since to rotation is counterclockwise by $\pi/6$ the square in (iv) is rotated to the 'straight' square. Similarly with the Fourier transform. The matches are with (iv) and (D).
5. $f\left(A^{-1}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)$. This is a clockwise rotation through $\pi/6$. The matches are with (i) and (E).
6. $f(x_1, x_2) * \operatorname{sinc}(ax_1)\operatorname{sinc}(ax_2)$. Well, there's only one choice left, so the matches have to be with (iii) and (B). Here's an explanation. Take the Fourier transform to get, by the convolution theorem,

$$\mathcal{F}(f(x_1, x_2) * \operatorname{sinc}(ax_1)\operatorname{sinc}(ax_2)) = \mathcal{F}f(\xi_1, \xi_2)\Pi_a(\xi_1)\Pi_a(\xi_2)\,.$$

This cuts off $\mathcal{F}f(\xi_1, \xi_2)$ by a 2D-rect function of width a. That matches with figure (B) (approximately – numerical computations, of course). In the spacial domain the effect is that of a lowpass filter, so the sharp edges of the square are smeared out somewhat. That's figure (iii).

(b) (i) Find the images $g_1(x,y)$, $g_2(x,y)$ and $g_3(x,y)$ using the Karhunen-Loeve (KL) transform. **[6]**

$$f_1(x,y) = \begin{cases} r_1 & 1 \le x \le \dfrac{M}{2}, 1 \le y \le M \\ r_2 & \dfrac{M}{2} < x \le M, 1 \le y \le M \end{cases}$$

Mean value of $f_1(x,y)$ is $m_1 = \dfrac{r_1}{2} + \dfrac{r_2}{2}$. Zero-mean version of $f_1(x,y)$ is

$$f_1(x,y) - m_1 = \begin{cases} \dfrac{r_1}{2} - \dfrac{r_2}{2} & 1 \le x \le \dfrac{M}{2}, 1 \le y \le M \\ \dfrac{r_2}{2} - \dfrac{r_1}{2} & \dfrac{M}{2} < x \le M, 1 \le y \le M \end{cases}$$

1

Digital Image Processing (DSP)

Mean value of $f_2(x,y)$ is r_3. Zero-mean version of $f_2(x,y)$ is $f_2(x,y)-m_2=0$.
Mean value of $f_3(x,y)$ is r_4. Zero-mean version of $f_3(x,y)$ is $f_3(x,y)-m_3=0$.
Variance of $f_1(x,y)-m_1$ is $\frac{1}{2}\frac{1}{4}(r_1-r_2)^2+\frac{1}{2}\frac{1}{4}(r_1-r_2)^2=\frac{1}{4}(r_1-r_2)^2$.
Variance of $f_2(x,y)-m_2$ is 0.
Variance of $f_3(x,y)-m_3$ is 0.
Covariance between $f_1(x,y)-m_1$ and $f_2(x,y)-m_2$ is 0. Therefore, the covariance matrix is $\begin{bmatrix} \frac{1}{4}(r_1-r_2)^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ with eigenvalues $\frac{1}{4}(r_1-r_2)^2$ and 0. Therefore, by using the Karhunen Loeve transform we produce three new images, with two of them being 0 and the other being $f_1(x,y)-m_1$.

(ii) Comment on whether you could obtain the result of (b)-(i) above using intuition rather than by explicit calculation. **[2]**

The above result is expected since two of the given images are constant and therefore they don't carry any information. This means that there is only one principal component in the given set.

2.

(i)

Yes

$$H = \begin{bmatrix} -1 & 2 & -1 \\ -2 & 4 & -2 \\ -1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 2 & -1 \end{bmatrix}$$

Horizontal filter is $[-1 \quad 2 \quad -1]$ and vertical filter is $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$.

[4]

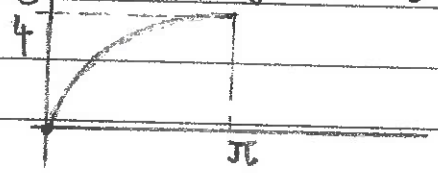
(ii)

We can write $h[m] = [-1 \quad 2 \quad -1]$ and

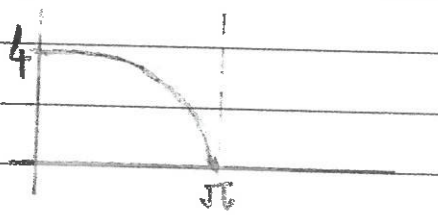
$h[n] = [1 \quad 2 \quad 1]$

In DTFT domain

$$H(u) = \mathcal{F}\{h[m]\} = 2 - e^{j\omega} - e^{-j\omega} = 2 - 2\cos u$$



$$H(v) = \mathcal{F}\{h[n]\} = 2 + 2\cos v$$



$$H(u,v) = 4(1-\cos u)(1+\cos v)$$

$$H(0,v) = 4(1-\cos 0)(1+\cos v) = 0$$

$$H(\pi,v) = 4(1-\cos\pi)(1+\cos v) = 8(1+\cos v)$$

$$H(u,0) = 8(1-\cos u)$$

$$H(u,\pi) = 0$$

[4]

(iii)

The filter is lowpass (smoothing) in vertical direction and highpass (sharpening) in horizontal direction. It can be used for detecting vertical edges.

[4]

(b) (i)

Pdf on the left:

The height is x

The area under the curve is 1.

Therefore, $64x + \frac{64x}{2} = 1 \Rightarrow x = \frac{1}{96}$

Transformation that yields an equalized image is:

$$T(r) = \int_0^r p_r(w)\,dw$$

The negative-slope line is described by the equation

$y = ax + \beta$

$x = 64, y = \frac{1}{96}, \quad x = 128, y = 0$

$\Rightarrow 64a + \beta = \frac{1}{96}$

$128a + \beta = 0$

From the above $a = -\frac{1}{64 \cdot 96} \quad \beta = \frac{2}{96}$

For $0 \le r \le 64$

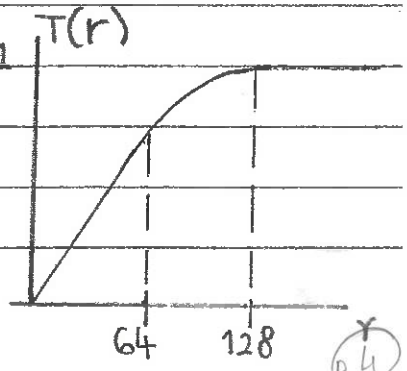
$$T(r) = \int_0^r p_r(w)\,dw = \frac{r}{96}$$

For $64 < r \le 128$

$$T(r) = \int_0^{64} p_r(w)\,dw + \int_{64}^r p_r(w)\,dw = \frac{1}{96} \cdot 64 + \int_{64}^r \left(-\frac{1}{64 \cdot 96} w + \frac{2}{96}\right) dw$$

$$= \frac{64}{96} + \frac{64^2}{2 \cdot 64 \cdot 96} - \frac{r^2}{2 \cdot 64 \cdot 96} + \frac{2r}{96} - \frac{2 \cdot 64}{96}$$

$$= -\frac{32}{96} + \frac{2r}{96} - \frac{r^2}{2 \cdot 64 \cdot 96}$$



DSP

[6]

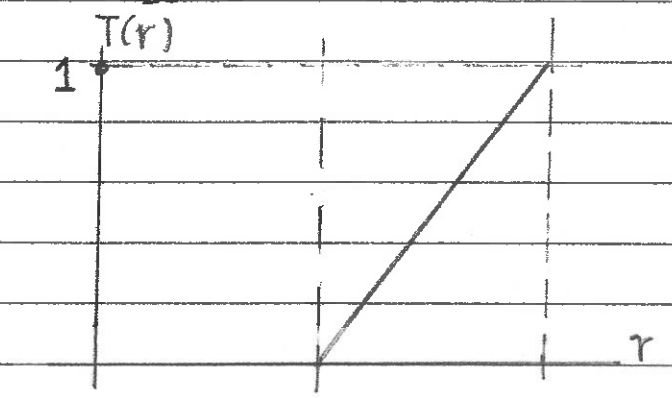
(ii)

Height is $\frac{1}{127}$

$T(r) = 0, \quad 0 \leq r < 128$

For $128 \leq r \leq 255$

$$T(r) = \frac{1}{127} \int_{128}^{r} dw = \frac{1}{127}(r - 128)$$



[2]

3 (a)
(i)

The objective is to minimize $J(\mathbf{f}) = \|\mathbf{n}(\mathbf{f})\|^2 = \|\mathbf{y} - \mathbf{H}\mathbf{f}\|^2$.

We set the first derivative of the cost function equal to zero $\frac{\partial J(\mathbf{f})}{\partial \mathbf{f}} = 0 \Rightarrow -2\mathbf{H}^{\mathrm{T}}(\mathbf{y} - \mathbf{H}\mathbf{f}) = \mathbf{0}$.

By solving that system we obtain $\mathbf{H}^{\mathrm{T}}\mathbf{H}\mathbf{f} = \mathbf{H}^{\mathrm{T}}\mathbf{y}$.

If $M = N$ and $\mathbf{H}^{-1}$ exists then $\mathbf{f} = \mathbf{H}^{-1}\mathbf{y}$.

According to the previous analysis if $\mathbf{H}$ (and therefore $\mathbf{H}^{-1}$) is block circulant the above problem can be solved as a set of $M \times N$ scalar problems as follows

$$F(u,v) = \frac{H^*(u,v)Y(u,v)}{|H(u,v)|^2} \Rightarrow f(i,j) = \Im^{-1}\left[\frac{H^*(u,v)Y(u,v)}{|H(u,v)|^2}\right] = \frac{Y(u,v)}{H(u,v)}$$

Erratic behaviour

Case 1

Suppose first that the additive noise $n(i,j)$ is negligible. A problem arises if $H(u,v)$ becomes very small or zero for some point (u,v) or for a whole region in the (u,v) plane. In that region inverse filtering cannot be applied. Note that in most real applications $H(u,v)$ drops off rapidly as a function of distance from the origin. **Solution:** if these points are known they can be neglected in the computation of $F(u,v)$.

Case 2

In the presence of external noise we have that

$$\hat{F}(u,v) = \frac{H^*(u,v)\big(Y(u,v) - N(u,v)\big)}{|H(u,v)|^2} = \frac{H^*(u,v)Y(u,v)}{|H(u,v)|^2} - \frac{H^*(u,v)N(u,v)}{|H(u,v)|^2} \Rightarrow$$

$$\hat{F}(u,v) = F(u,v) - \frac{N(u,v)}{H(u,v)}$$

If $H(u,v)$ becomes very small, the term $N(u,v)$ dominates the result.

Solution: again to carry out the restoration process in a limited neighborhood about the origin where $H(u,v)$ is not very small. This procedure is called **pseudoinverse filtering**. In that case we set

$$\hat{F}(u,v) = \begin{cases} \dfrac{H^*(u,v)Y(u,v)}{|H(u,v)|^2} & H(u,v) \neq 0 \\ 0 & H(u,v) = 0 \end{cases} \quad \text{or} \quad \hat{F}(u,v) = \begin{cases} \dfrac{H^*(u,v)Y(u,v)}{|H(u,v)|^2} & |H(u,v)| \geq \varepsilon \\ 0 & \text{otherwise} \end{cases}$$

In general, the noise may very well possess large components at high frequencies (u,v), while $H(u,v)$ and $Y(u,v)$ normally will be dominated by low frequency components. The parameter ε (called **threshold**) is a small number chosen by the user.

(ii)
In a particular scenario the degradation process can be modelled as a linear filter with the two-dimensional impulse response given below:

$$h(x,y)=\begin{cases}0.25 & x\pm 1, y=0\\ 0.5 & x=0, y=0\\ 0 & \text{elsewhere}\end{cases}$$

Inverse Filtering cannot be applied if $H(u,v)=0$.

$$H(u,v)=\frac{1}{NN}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1}h(x,y)e^{-j2\pi(ux/N+vy/N)}=\frac{0.25}{NN}(e^{-j2\pi u/N}+e^{j2\pi u/N}+2)$$

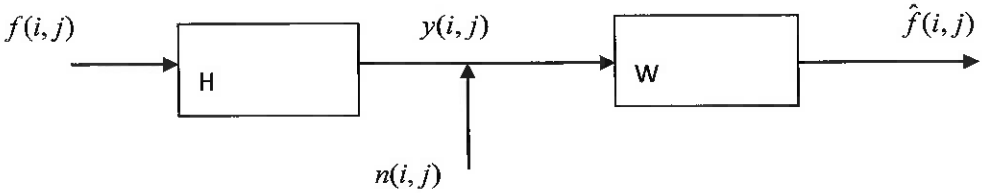
$$=\frac{0.5}{NN}[\cos(2\pi u/N)+1]$$

The values of (u,v) for which $H(u,v)$ cannot be estimated are found by:

$$\cos(2\pi u/N)+1=0\Rightarrow\cos(2\pi u/N)=-1\Rightarrow u=\pi$$

(b)

The image restoration problem can be viewed as a system identification problem as follows:



The objective is to minimize the following function (least squares error filter)

$$E\{(\mathbf{f}-\hat{\mathbf{f}})^{\mathrm{T}}(\mathbf{f}-\hat{\mathbf{f}})\}$$

To do so the following conditions should hold:

1. $E\{\hat{\mathbf{f}}\}=E\{\mathbf{f}\}\Rightarrow E\{\mathbf{f}\}=\mathbf{W}E\{\mathbf{y}\}$
2. the error must be orthogonal to the observation about the mean

   $E\{(\hat{\mathbf{f}}-\mathbf{f})(\mathbf{y}-E\{\mathbf{y}\})^{\mathrm{T}}\}=0$

In frequency domain

$$W(u,v)=\frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)|H(u,v)|^2+S_{nn}(u,v)},\quad \hat{F}(u,v)=\frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)|H(u,v)|^2+S_{nn}(u,v)}Y(u,v)$$

Computational issues

The noise variance has to be known, otherwise it is estimated from a flat region of the observed image. In practical cases where a single copy of the degraded image is available, it is quite common to use $S_{yy}(u,v)$ as an estimate of $S_{ff}(u,v)$. **This is very often a poor estimate.**

Wiener smoothing filter

In the absence of any blur, $H(u,v)=1$ and

$$W(u,v)=\frac{S_{ff}(u,v)}{S_{ff}(u,v)+S_{nn}(u,v)}=\frac{(SNR)}{(SNR)+1}$$

(a) $(SNR)\gg 1 \Rightarrow W(u,v)\cong 1$

(b) $(SNR)\ll 1 \Rightarrow W(u,v)\cong (SNR)$

(SNR) is high in low spatial frequencies and low in high spatial frequencies so $W(u,v)$ can be implemented with a lowpass (smoothing) filter.

Relation with inverse filtering

If $S_{nn}(u,v)=0 \Rightarrow W(u,v)=\frac{1}{H(u,v)}$ which is the inverse filter

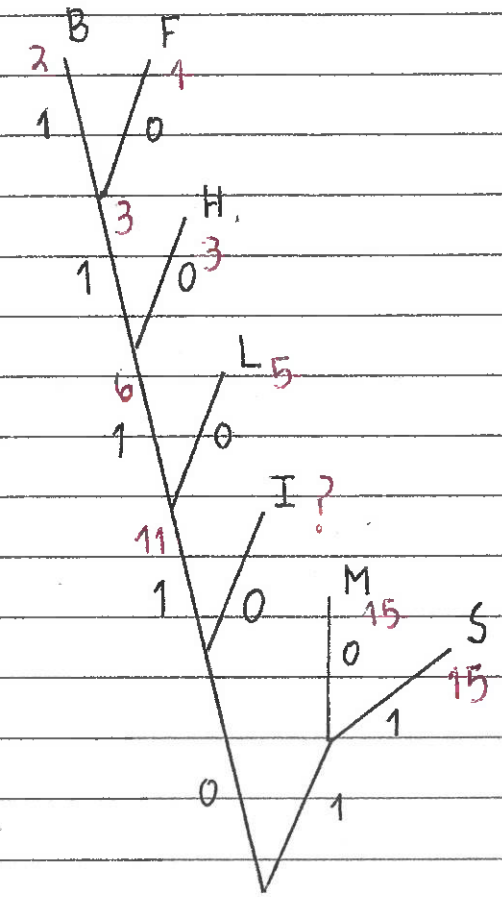
If $S_{nn}(u,v)\to 0$

$$\lim_{S_{nn}\to 0} W(u,v)=\begin{cases}\dfrac{1}{H(u,v)} & H(u,v)\neq 0\\ 0 & H(u,v)=0\end{cases}$$

which is the pseudoinverse filter.

4(a)

(i)



[5]

(ii)

We see that $? \leq 15$ and furthermore

$11 + ? \geq 15 \Rightarrow ? \geq 4 \Rightarrow$

$4 \leq ? \leq 15$

1 2 3 (4) (5) (6) (7) (8) (9) (10) (11) (12) (13) (14) (15) 16 17

18 19 20

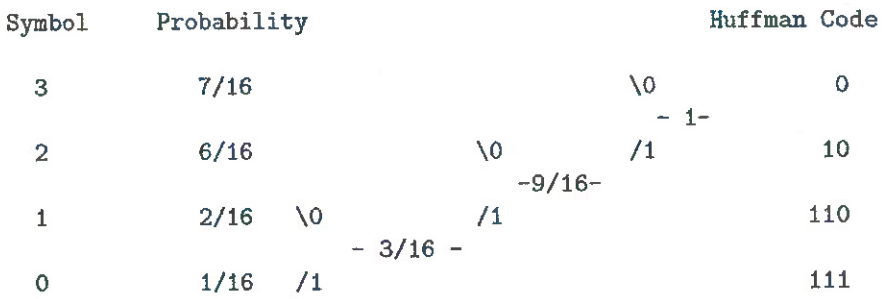
[5]

SOLUTIONS

(b)

(i)

Huffman Coding. Calculate probability of each symbol.



Mean bits/pixel $1 \times 7/16 + 2 \times 6/16 + 3 \times 2/16 + 3 \times 1/16 = 28/16 = 1.75$ bits/pixel. Plain message required 2 bits/pixel. Compression ratio achieved is $2/1.75 = 1.14$

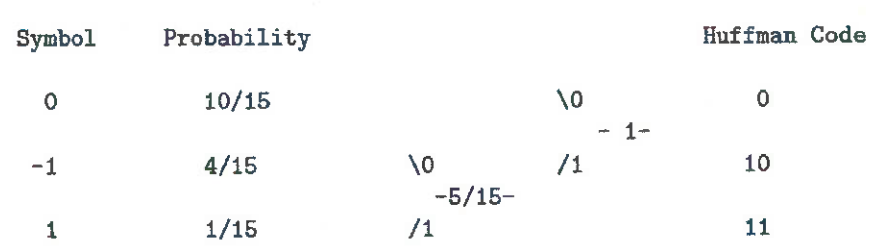
[5]

(ii)

Form differences via horizontal raster scan

	0	0	-1
0	1	0	0
0	-1	0	0
0	-1	0	-1

Calculate Huffman code.



Number of bits to transmit image $1 \times 10 + 2 \times 4 + 2 \times 1 = 20$ but must add 2 bits to transmit first undifferenced value in image top left corner, hence total of 22 bits needed. Plain message has 16 pixels with 2bits/pixel = 32bits. Compression ratio is therefore $32/22 = 1.45$.

[5]